

Bill of Materials — T8-40g

VBF-T8R1-BOM-01 Case t8_r1 — Long-endurance drone battery 2026-08-25

Prepared: Reviewed: Approved: _____

BOM

Component	g/cell	kg/kWh	Formula / source
Positive electrode active material (NMC811)	16.4465	0.6515	positive layer mass × 0.96 wt-frac (lit. default)
Positive electrode conductive additive (CB)	0.3426	0.0136	positive layer mass × 0.02 wt-frac (lit. default)
Positive electrode binder (PVDF)	0.3426	0.0136	positive layer mass × 0.02 wt-frac (lit. default)
Negative electrode active material (graphite)	12.3423	0.4889	negative layer mass × 0.955 wt-frac (lit. default)
Negative electrode conductive additive (CB)	0.1292	0.0051	negative layer mass × 0.01 wt-frac (lit. default)
Negative electrode binder (SBR/CMC)	0.4523	0.0179	negative layer mass × 0.035 wt-frac (lit. default)
Separator	0.2257	0.0089	separator layer_kg_m2 × area (contract formula)
Electrolyte (not in contract mass)	11.0298	0.4369	pore volume × 1.2 g/cm3 (literature, annotated)
Positive current collector Al	2.7283	0.1081	8 μm × 2700 kg/m3 × area (contract formula)
Negative current collector Cu	6.7904	0.269	6 μm × 8960 kg/m3 × area (contract formula)
Enclosure / tabs			Not modeled
Total (contract caliber, excl. electrolyte)	39.8	1.5765	Σ layer rows above; electrolyte excluded (set lacks density)
Total including electrolyte (lit. density)	50.8298	2.0134	informational (adds electrolyte row above)
Cell energy	25.246	Wh	r5_final_energy.json (√V·I dt)